IEEE TAMU TECHNOTHON · ENGINEERING FOR DISASTERS

AFTERSHOCK

NoStack Developers

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What we’re building

THE PROBLEM

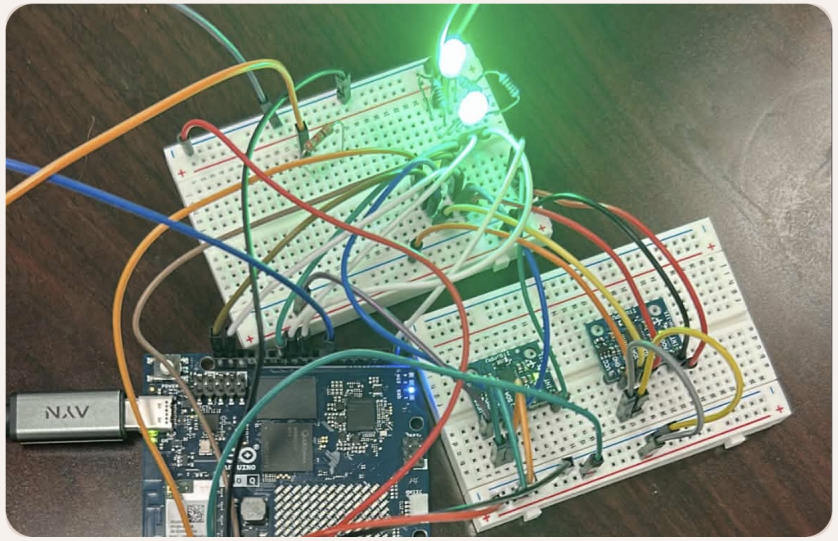
After an earthquake, the danger isn’t over.

Aftershocks can shift a damaged building after rescuers and residents have gone back in, and there’s rarely a simple way to know when to get out.

OUR PROJECT

1. **Two motion sensors** mounted on a structure feel every shake.
2. **An Arduino UNO Q** decides on the spot whether it’s a real aftershock, and how serious.
3. **Everyone sees the risk** in plain colors: a light, a screen, a live dashboard and a College Station simulation.

The brain: Arduino UNO Q





Close-up: the STM32U585 microcontroller, on the back of our board.

REAL-TIME MICROCONTROLLER

STM32U585

- Reads both motion sensors 50 times a second
- Drives the RGB light and the LCD screen
- Never pauses, so no sample is missed

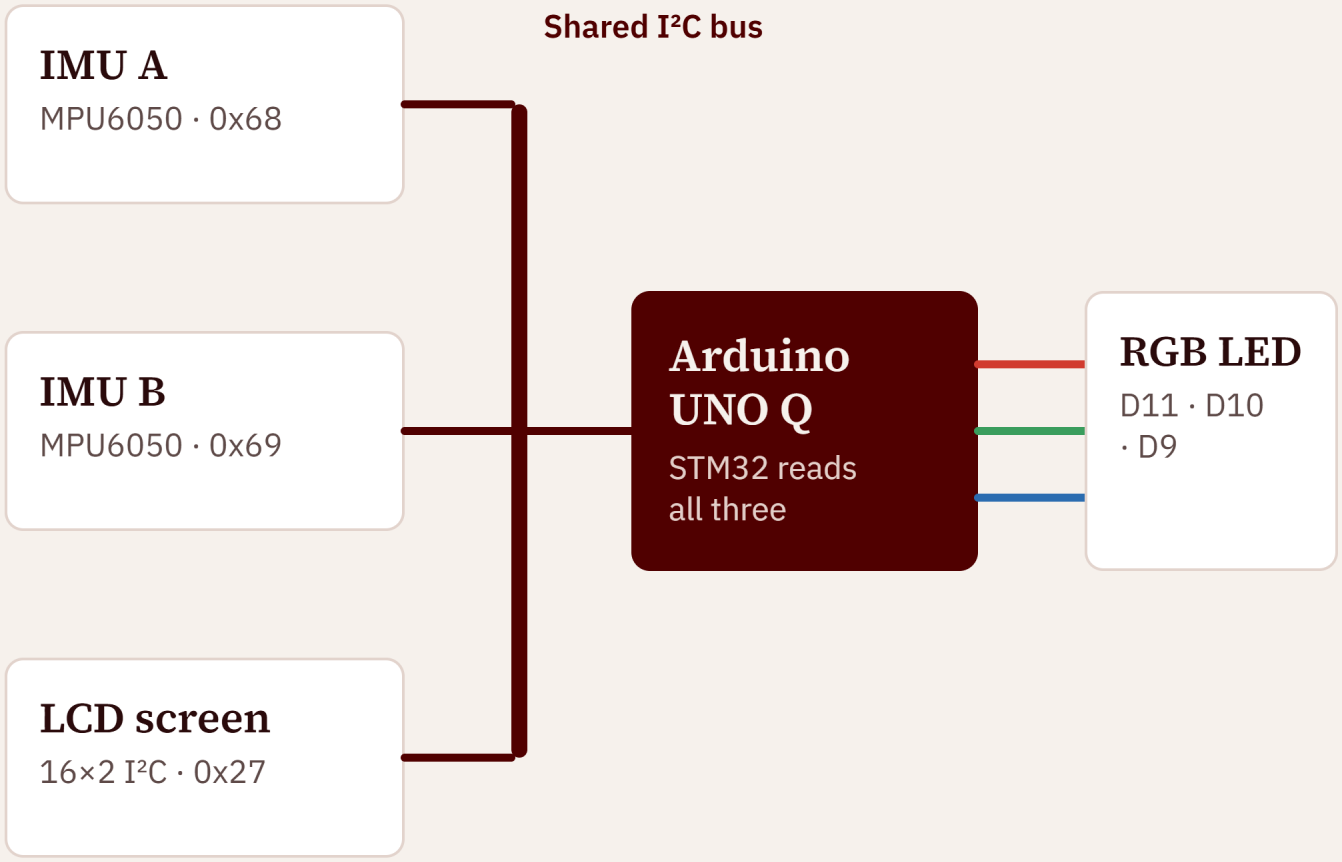
Bridge: sensor data up, risk level back down

LINUX COMPUTER

Qualcomm QRB2210

- Runs Model 1, our motion classifier, in Python
- Hosts the live dashboard over Wi-Fi
- 2–4 GB RAM, dual-band Wi-Fi 5

The circuit



Part	Pin	Goes to
IMU A	AD0	GND → address 0x68
IMU B	AD0	3.3 V → address 0x69
IMUs + LCD	SDA, SCL	UNO Q SDA, SCL
IMUs	VCC, GND	3.3 V, GND
RGB LED	R, G, B	D11, D10, D9 via 220–330 Ω

Off-the-shelf parts. Two sensors and a screen share the same two data wires.

Model 1: feeling a real shake

It **calibrates** to the room for 8 seconds, then turns each second of motion into a **16-number embedding** built from the **peak** (strongest jolt) and **RMS** (sustained energy).

Risk shown on the light and screen

Baseline | **Low** | **Medium** | **High**

Model 2: seeing it across College Station

A machine-learning model rates each district, and the 3D twin shows the result.

6 GROUND-MOTION INPUTS

Input	What it captures
PGA	Strongest single jolt
Arias intensity	Total shaking energy
Duration	How long strong shaking lasts
CAV	Damage-related build-up
Soil factor	How much the ground amplifies
Building age	How fragile the district is

Random forest

13 decision trees each judge the inputs, and their votes are averaged.

Trained on [training data source]

- **Nominal: open**
- **Advisory: caution**
- **Lockdown: gates close**

In the twin, districts change color and crossings such as the Joe Routt rail crossing close, so anyone can see the risk without reading a number.